ETHUSDT / 15-MINUTE HORIZON

Strategy and Rules Report

What the local dashboard actually uses to estimate a target touch, what was tested, and what was rejected.

> **Scope:** This report documents the code in `stock_details.html`, `indicators.js`, `backtest.py`, `walkforward_results.json`, and the Pine indicators in this workspace.
> **Instrument:** ETHUSDT. **Live source:** Binance Vision through the local Python proxy. **Base data:** 1-minute OHLCV candles.

> **Important:** This is decision-support software, not a guaranteed prediction, trading signal, investment recommendation, or execution system. A probability means historical frequency under similar conditions; it does not mean the target must be reached.

Prepared from the implemented repository logic. The live dashboard refreshes price every 5 seconds and refreshes its 500-candle history approximately every 60 seconds while streaming.

1. Executive Summary

The dashboard does not use a neural network or a deterministic forecast price. Its primary live engine estimates the probability that ETH will **touch a user-entered target level** within each minute from the current point through the end of the current 15-minute window.

1. Fetch the current ETHUSDT price.
2. Fetch up to 500 recent 1-minute candles.
3. Measure recent momentum, current volatility, recent volume, and RSI(14).
4. Search historical candles for similar momentum and volatility conditions.
5. Measure whether the requested percentage move was touched within 1 to 15 minutes after those historical points.
6. Weight recent historical examples more heavily, blend matched examples with all examples, and calibrate the resulting probability.
7. Display a minute-by-minute probability, median move, 90th-percentile move, and a simple Can Cross decision.

> **What is live:** target-crossing probability, historical move statistics, trend label, volatility regime, volume label, RSI value, and a gated RSI reversion label.

> **What is not live:** the EMA9/EMA21 + MACD strategy, Bollinger strategy, raw momentum bias, Pine Script alerts, order execution, stop-loss management, position sizing, and news/sentiment analysis.

2. Data and Timing Rules

Item	Implemented rule
Symbol	ETHUSDT.
Input candles	Up to 500 recent 1-minute Binance Vision klines. Each candle contains open, high, low, close, and volume.
Prediction horizon	1 through 15 minutes. The 15-minute row represents a touch at any point during the next 15 one-minute candles.
Target definition	For an UP target: target distance = target - current price. For a DOWN target: current price - target. The distance is converted to a percentage of current price.
Historical outcome	For each historical base close, use the maximum future high excursion for UP or maximum future low excursion for DOWN over the selected horizon.
Lookahead control	The live calculation uses only already-fetched history. The causal backtest excludes historical outcomes that would not yet have resolved at the simulated time.
Refresh	Price and current-window volume refresh every 5 seconds. The 500-candle history refreshes every 12 ticks, approximately once per minute.
Time basis	Countdowns use UTC interval boundaries.

3. Primary Live Strategy: Historical Target-Touch Probability

3.1 Recent momentum classification

The dashboard computes the return from the open of the oldest candle in the last 20 candles to the close of the newest candle:

```
momReturn = (last close - first open) / first open * 100
trend = UP if momReturn > 0.02%
       DOWN if momReturn < -0.02%
       FLAT otherwise
```

This trend is descriptive context. It is not itself the final target probability.

3.2 Volatility regime

For every eligible historical point, the code computes a 20-bar range-based ATR percentage:

```
ATR20(i) = average over 20 bars of ((high - low) / close) * 100
ATR ratio = current ATR20 / average historical ATR20
Hot    if ATR ratio >= 1.5
Normal if 0.8 <= ATR ratio < 1.5
Quiet  if ATR ratio < 0.8
```

For a matched historical sample, its ATR must be between 0.5x and 2.0x the current ATR. This prevents comparing an unusually calm historical period with an unusually volatile current period, or vice versa.

3.3 Historical matching

For each horizon `m = 1..15`, every historical point after the 20-bar warmup is evaluated. A point is a matched sample when:

- Its 20-bar momentum label equals the current trend label.
- Its ATR is between 0.5x and 2.0x current ATR.

The matched sample set is used only when it contains at least 15 observations. Otherwise the model falls back to all eligible historical samples.

3.4 Recency weighting

Recent examples receive exponentially higher weight:

```
weight(i) = exp(-0.006 * (latest index - historical index))
```

```
The matched probability receives 70% weight and the all-history probability receives 30% weight. Wit
```

3.5 Calibration

```
The raw historical estimate is passed through the piecewise-linear calibration curve in indicators.j
```

Raw estimate	Displayed estimate	Raw estimate	Displayed estimate
0.0%	0.0%	55.0%	55.3%
3.7%	7.6%	65.1%	63.3%
15.2%	20.5%	75.0%	72.9%
34.9%	34.6%	83.7%	74.7%
45.1%	44.0%	92.1%	77.1%
100.0%	77.1%	Linear interpolation between points	

4. What Each Projection Table Column Means

Column	Meaning
Min	Minute number inside the current 15-minute window.
Countdown	Time remaining until that minute boundary, based on U
Median Move	Median historical favorable excursion among the selec

90th %ile	90th-percentile favorable excursion among the selecte
Can Cross?	YES if calibrated probability is at least 30%, or if
Probability	Calibrated chance that the target level is touched du
Status	Very likely >=70%, Likely >=50%, Possible >=30%, Low
Volume	Current-window 1-minute candle direction and raw volu

> **Touch versus close:** A wick through the target counts as a hit. The model does not require a candle c

5. Secondary Live Context Signal: RSI(14) Reversion

The dashboard calculates causal Wilder RSI(14) from the 1-minute closes:

```
RSI = 100 - 100 / (1 + average gain / average loss)
UP candidate  if RSI < 30
DOWN candidate if RSI > 70
```

The candidate is only displayed as a signal when the volatility regime is **Normal**. In Hot or Quiet re

Condition	Displayed label
RSI < 30 and Normal volatility	UP, with a hard-coded historical accuracy context of
RSI > 70 and Normal volatility	DOWN, with a hard-coded historical accuracy context
RSI between 30 and 70	No signal.
Extreme RSI outside Normal volatility	No signal; regime deemed unreliable.

The RSI signal does not replace the target-touch probability. It is a separate mean-reversion contex

6. Strategies Tested in Backtests

6.1 RSI(14) extreme reversion

At RSI below 30, predict UP; above 70, predict DOWN. The gated version only permits calls in the Nor

6.2 EMA9/EMA21 + MACD trend confirmation

Compute EMA(9), EMA(21), and MACD(12,26,9). The walk-forward implementation predicts UP when EMA9 >

6.3 Bollinger Bands(20,2) mean reversion

Compute a 20-close simple moving average and bands at two population standard deviations. A close ab

6.4 Historical momentum-only bias

The older bias experiment matched the current 20-bar momentum label to historical labels and estimat

6.5 Baselines

Walk-forward comparison includes Always UP, Always DOWN, and deterministic Random 50%. These are san

7. Walk-Forward Results in the Repository

The saved report uses ten fixed, non-overlapping windows. Each window contains approximately 3,000 o

Strategy	Calls / coverage	Accuracy	Net expectancy after C	Decision
Always UP	29,450 / 100.0%	49.54%	-0.2929%	Baseline only
Always DOWN	29,450 / 100.0%	50.46%	-0.3071%	Baseline only
Random 50%	29,450 / 100.0%	50.04%	-0.3000%	Baseline only
RSI raw	2,480 / 8.42%	55.04%	-0.3296%	Not promoted
RSI + Normal ATR	1,058 / 3.59%	57.28%	-0.2921%	Not promoted
EMA/MACD	19,317 / 65.59%	49.23%	-0.2918%	Rejected

Promotion result: DO NOT PROMOTE. The candidate needed positive net expectancy after costs, a 95% co

The 0.30% round-trip assumption is explicitly modeled as 0.10% taker fee plus 0.05% slippage per sid

8. Pine Script Indicators in This Workspace

These scripts are separate TradingView indicators. They are not called by the local Python server or

Script	Rules represented	Status in local dashboard
eth_signals.pine	EMA9, EMA21, VWAP, RSI14; BUY when above VWAP, EMA9 > EMA21, RSI 50–70, volume > 1.2× 20-bar average	Separate TradingView indicator
eth_event_master.pine	EMA9/21, VWAP, RSI14, Stoch RSI, Bollinger Bands, volume filters,	Separate TradingView indicator
eth_event_master_v2.pine	Expanded event dashboard with EMA/VWAP/RSI/Stoch RSI, Bollinger	Separate TradingView indicator
eth_event_master_v3.pine	Streamlined version using EMA9/21, VWAP, RSI14, Stoch RSI, take-prof	Separate TradingView indicator

9. Limitations and Correct Interpretation

- **Probability is conditional frequency:** it estimates historical target touches under approximate si
- **Small target distance can look easy:** the target percentage is measured from the live price and is
- **Wicks count:** a brief high or low touching the level is a success in the model.
- **Sample size matters:** the table marks thin samples below 50 as less reliable.
- **Regime drift:** crypto behavior can change; a 500-minute history is not a complete market model.
- **No execution controls:** there is no order placement, stop, take-profit order, position sizing, lev
- **No news model:** the local assistant does not currently fetch or interpret live news or external se
- **Backtest cost sensitivity:** the reported candidate returns are negative after the chosen cost assu
- **Not financial advice:** any use should include independent validation, defined risk, and a willingn

10. Practical Reading Checklist

1. Confirm the current price and target direction.
2. Check the target distance percentage. A farther target requires a larger historical excursion.
3. Check whether trend agrees with the target direction.
4. Check volatility regime. Hot and Quiet suppress the RSI reversion label.
5. Prefer rows with adequate samples; treat thin samples cautiously.
6. Read “Can Cross?” as a historical possibility filter, not an instruction to trade.
7. Use a predefined risk limit and do not rely on the dashboard alone.

Source implementation: ETH Projection local dashboard. Report generated from repository logic on 2026-09-2